

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B1: FLUIDS

TRINITY TERM 2024

Friday, 7 June, 2:30 pm – 4:30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

The Navier–Stokes equation for viscous, incompressible fluid flow under gravity is

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} + \frac{1}{\rho} \nabla p + g \hat{\mathbf{z}} = \nu \nabla^2 \mathbf{v} ,$$

where $\mathbf{v}$ is the fluid velocity, ρ the density, p the pressure, g the acceleration due to gravity, $\hat{\mathbf{z}}$ the vertical unit vector and ν the kinematic viscosity.

$$1000 \text{ mb} = 1000 \text{ hPa} = 10^5 \text{ Pa}.$$

1. (a) Define what is meant by the *stream function* and *vorticity* in a two-dimensional fluid flow, and write down the equation relating these quantities. Show that the stream function

$$\psi = -\frac{\Omega(x^2 + y^2)}{2},$$

corresponds to a fluid moving in solid body rotation with rotation rate Ω . Calculate the vorticity in this flow. [7]

(b) An elliptical tank with a solid wall at $(x/L_x)^2 + (y/L_y)^2 = 1$ is filled with water and placed on a rotating turntable. The turntable is rotated at a rate Ω and the water within the tank brought into solid body rotation. Sketch the streamlines for the flow as observed in the non-rotating frame of the laboratory and explain why these streamlines intersect the walls of the tank. [4]

(c) The rotation of the turntable is halted abruptly. State the value of the vorticity within the subsequent fluid flow, assuming the flow is inviscid, stating how you arrive at your answer. Hence verify that

$$\psi = Ax^2 + By^2$$

is a solution for the stream function, and determine the values of A and B . [8]

(d) Sketch the streamlines for the flow in part (c) and describe the motion of a paddle wheel placed into the flow in the interior of the tank. Sketch how, and explain physically why, the motion of a paddle wheel differs adjacent to the wall of the tank in a fluid at high Reynolds number. [6]

2. (a) Write down the equation relating the velocity, $\mathbf{v}$, to the velocity potential, ϕ , in an irrotational flow. Derive and interpret physically the condition under which the latter satisfies Laplace's equation,

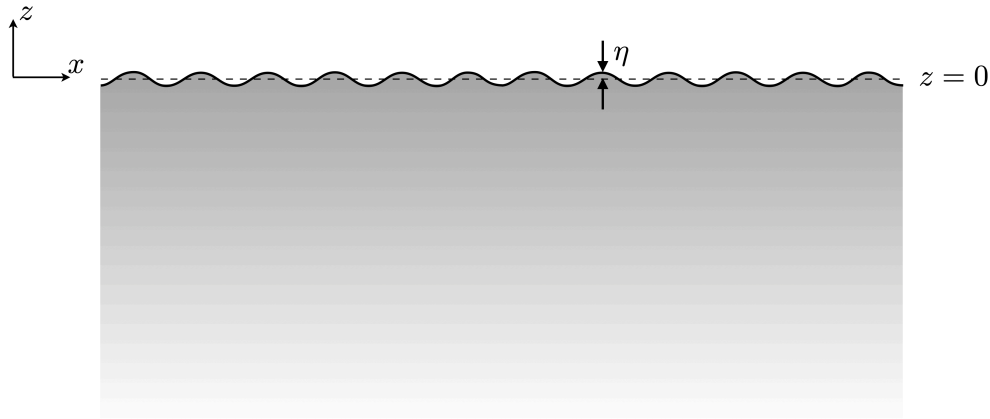
$$\nabla^2 \phi = 0.$$

Assuming, additionally, that the flow is inviscid and the density is uniform, show that

$$\frac{\partial \phi}{\partial t} + \frac{\nabla \phi \cdot \nabla \phi}{2} + \frac{p}{\rho} + gz = C,$$

where C is a constant of integration. Describe in a few sentences the physical significance of the terms in this equation. [8]

(b) Consider surface waves in the ocean of wavelength much shorter than the ocean depth, as sketched in the figure. The elevation of the free surface is $\eta = A \cos(kx - \omega t)$ and you can treat the density as uniform.



Show that for waves of small amplitude, the surface kinematic boundary condition is well approximated by

$$\frac{\partial \phi}{\partial z} = \omega A \sin(kx - \omega t) \quad \text{at } z = 0.$$

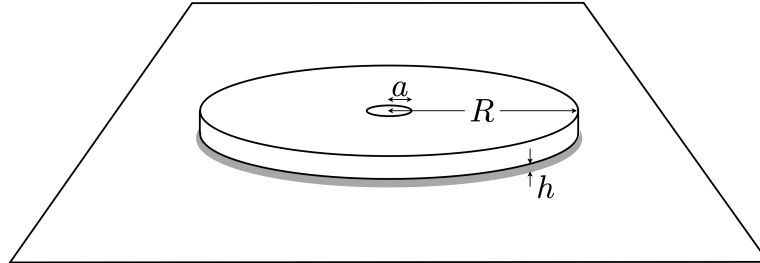
Write down and justify a further boundary condition satisfied at the surface for waves of small amplitude. Obtain the solution for ϕ and show that the waves satisfy the dispersion relation

$$\omega = \sqrt{gk}.$$

How are the phase and group speeds of the waves related? Explain, with a suitable sketch, what this means for the evolution of a wave packet. [12]

(c) Write down expressions for v_x and v_z at the sea surface. Hence sketch the nature of the motion of a fluid parcel at the sea surface with time, assuming that the wave is of small amplitude and has positive phase velocity. What is the value of the vorticity of a fluid parcel at the sea surface, and why? [5]

3. A disk of radius R with a small hole at $r \leq a \ll R$ is separated by a viscous fluid of thickness $h(t) \ll R$ from a smooth horizontal table at $z = 0$, as sketched in the figure. The velocity has radial and vertical components v_r and v_z respectively. The disk is raised slowly from the table.



(a) Starting from the Navier Stokes equation and clearly stating any assumptions, derive the thin film equation that describes the flow of the fluid between the disk and the table. Write down the boundary conditions for the radial velocity at $z = 0$ and $z = h(t)$. Hence show that the radial velocity is given by

$$v_r = -\frac{1}{2\rho\nu} \frac{\partial p}{\partial r} z(h-z).$$

[6]

(b) State the physical origin of the equation

$$\frac{1}{r} \frac{\partial}{\partial r} (rv_r) + \frac{\partial v_z}{\partial z} = 0.$$

Derive an expression for the vertical velocity at $z = h(t)$. Hence, or otherwise, show that pressure is given by

$$p = p_0 - \frac{3\rho\nu}{h^3} \left[(R^2 - r^2) - (R^2 - a^2) \frac{\ln(r/R)}{\ln(a/R)} \right] \frac{dh}{dt},$$

where p_0 is a constant, stating clearly any assumptions you make.

[11]

(c) Explain why a mechanical force must be applied to lift the disk from the table. A disk of radius 1 m with no hole at its centre, i.e., for which $a = 0$, is separated from a smooth horizontal table by a thin film of water of uniform thickness 0.5 mm. Calculate the force required to lift the disk at a rate of 1 mm s^{-1} , given that the density and kinematic viscosity of water are 10^3 kg m^{-3} and $10^{-6} \text{ m}^2 \text{ s}^{-1}$ respectively. Explain, without detailed calculation, the impact of introducing a small hole on the force required to lift the disk.

[8]

4. (a) The equations for adiabatic, inviscid flow in one dimension are

$$\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = 0,$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho v_x) = 0,$$

$$\frac{p}{p_0} = \left(\frac{\rho}{\rho_0} \right)^\gamma,$$

where $\gamma = 7/5$ and p_0, ρ_0 are constants. By linearising these equations about a reference state of rest and uniform density, derive the wave equation for linear sound waves in the absence of any background flow. Show that both the phase and group speed are given by

$$c = \sqrt{\frac{dp}{d\rho}}.$$

Estimate the speed of sound waves at surface atmospheric pressure of 10^5 Pa and air density of 1.2 kg m^{-3} . [13]

- (b) The quantity P is defined such that

$$dP = \frac{dp}{\rho c}.$$

Use this definition to derive the coupled nonlinear sound wave equations:

$$\left[\frac{\partial}{\partial t} + (v_x + c) \frac{\partial}{\partial x} \right] (v_x + P) = 0;$$

$$\left[\frac{\partial}{\partial t} + (v_x - c) \frac{\partial}{\partial x} \right] (v_x - P) = 0.$$

Consider the solution in which $v_x = P$, such that the second equation is satisfied trivially. By interpreting the first equation as a material derivative, infer and sketch how the sound wave evolves with time if $v_x = v_0 \sin(kx)$ at time $t = 0$. Explain why the solution breaks down in a finite time, and estimate the time scale on which this break down occurs. [12]